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Review

Assessment and Monitoring of Groundwater Contaminants in Heavily Urbanized Areas: A Review of Methods and Applications for Philippines

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Abstract

Groundwater remains a critical water source for urban communities, particularly in rapidly urbanizing countries such as the Philippines. However, intensifying anthropogenic pressures have contributed to widespread contamination from heavy metals, nutrients, pathogens, volatile organic compounds (VOCs), and emerging pollutants, including pharmaceuticals and personal care products (PPCPs). This review synthesizes findings from 130 peer-reviewed studies on groundwater monitoring and remediation, emphasizing technological advancements and their application in urban environments. The literature is categorized into five thematic areas: monitoring technologies, contaminant profiles, remediation strategies, Philippine-specific case studies, and alignment with global frameworks. Recent innovations—such as Internet of Things (IoT)-enabled systems, remote sensing, biosensors, and artificial intelligence/machine-learning (AI/ML) models—show strong potential for real-time and predictive monitoring. Despite these advancements, technology adoption in the Philippines remains limited due to regulatory, technical, and infrastructural constraints. This review identifies key research and implementation gaps, particularly in the monitoring of emerging contaminants and the integration of data into policy-making and urban planning. To address these challenges, a conceptual framework is proposed to support more sustainable, technology-driven, and context-sensitive groundwater management in heavily urbanized areas.

Keywords: groundwater monitoring; urban contamination; IoT sensors; biosensors; Philippines

Academic Editor: Constantinos V. Chrysikopoulos

Received: 19 May 2025
Revised: 17 June 2025
Accepted: 23 June 2025
Published: 26 June 2025

Citation: Robles, K.P.V.; Monjardin, C.E.F. Assessment and Monitoring of Groundwater Contaminants in Heavily Urbanized Areas: A Review of Methods and Applications for Philippines. *Water* **2025**, *17*, 1903. https://doi.org/10.3390/w17131903

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1. Introduction

Groundwater is an essential resource from a global perspective and provides the largest store of freshwater, apart from the ice caps. Current groundwater abstraction represents approximately 26% of total freshwater withdrawal globally [1]. Since groundwater supplies almost half of all drinking water globally, it contributes to 43% of the global consumptive use in irrigation. In arid and semi-arid regions, groundwater is the only reliable water resource [2]. The efforts to preserve groundwater have become a necessity because of overuse and contamination of it. As evident globally, the quality of

groundwater has been severely polluted because of urbanization [3], industrialization [4,5], groundwater extraction [6], climate change [7], and poor disposal of waste [8,9]. People worldwide are still exposed to the dangers of contaminants in groundwater, and health concerns continue to escalate due to groundwater pollution [10].

In the Philippines, traces of groundwater contaminants in several areas have been investigated and studied. Natural and anthropogenic sources of pollutants have been found in the groundwater in some provinces: chromium, manganese, and nickel were found in groundwater near abandoned mines and vicinities in Puerto Princesa, which exceeded standards for drinking water [11]; 88.46% of groundwater samples contain high levels of arsenic in wells of unique locations in Batangas province [12]; the concentration of arsenic that is beyond drinking limit was discovered in Pampanga [13]; lead and cadmium in sanitary landfills in Cebu exceeded the national standard for drinking water [14]; nitrate contents in groundwater exceed the acceptable limits set by the World Health Organization (WHO) in Santa Ignacia, Tarlac [15]; and groundwater from wells near landfills was found to be unsafe for drinking because of total coliform and E. coli in La Union [16]. Moreover, a recent study explained the insufficiency of existing programs and policies in the early detection of arsenic in the Philippines. These studies show that groundwater pollutants are becoming a growing concern not only in the Philippines but also in other countries around the world [17]. Thus, there is a necessity for regular monitoring and detection of contaminants in the groundwater to determine the best mitigation and effective regulations for reducing and mitigating groundwater pollution.

Nations have been relying on groundwater as their primary potable water source, as one-third of the population globally relies on groundwater [18]. Additionally, groundwater was considered to have contributed to human development [19]. However, the rise in population will increase the water demand, which poses a risk to groundwater availability and quality. As the problems in quantity and quality of groundwater become a threat, the enhancement in monitoring and remediation of it is necessary to develop groundwater contaminant management strategies and ensure the preservation and availability of groundwater resources [7]. Pollution is one of the reasons for water scarcity, and addressing this problem can provide people with safe and sustainable water. It would meet one of the sustainable goals of the United Nations by 2030, which is to ensure the availability and sustainable management of water and sanitation for all [20].

It is crucial to address serious environmental problems associated with contaminated water. Both naturally occurring and man-made groundwater contamination provide major health and environmental dangers, with organic contaminants, pesticides, and heavy metals being the most dangerous. Case studies from nations like Ethiopia, China, and India highlight the problem's worldwide reach and the demand for creative remediation and detection methods, urging collaboration across nations to address these concerns [20]. Comparably, new pollutants are showing up in the environment more frequently. These include pesonal care items and medications, and their permanence, as well as potential health risks, raises worries. In order to solve these issues, the assessment draws attention to the gaps in international rules, particularly in poor nations, and calls for stronger laws and improved environmental monitoring [21].

Furthermore, groundwater is said to be an invisible natural resource, and because of this, the pollution on it goes unnoticed. In the hydrologic cycle, water is filtered from the soil and flows back to streams and rivers or extracted for consumption [21]. However, the potential for groundwater resources is becoming limited due to increased groundwater pollution. Pollution may emerge from point and nonpoint sources, which can be difficult to investigate [22]. Sidiropoulos (2023) states that the sources, behavior, and transport of pollutants are essential to study to be able to identify, measure, and control them [22]. On the other hand, efforts to monitor and identify pollutants in groundwater have been

studied and researched. A review article provides methods for identifying groundwater pollution sources and remediation technologies [23]. The authors suggest using combination technologies in pollution source detection to address complex problems and improve the accuracy in identifying the source of pollution. The critical thing to do is to identify pollution sources to provide accurate information about pollution sources, which is a prerequisite to developing efficient remediation strategies [23]. Furthermore, groundwater quality monitoring is one solution for identifying and monitoring pollution and providing details on pollutants. With the advancement of technologies in groundwater monitoring [19], faster response to pollution would be expected, and groundwater pollution will potentially be eradicated.

To contribute to the growing concern, this review aims to present and evaluate groundwater quality monitoring and mitigation studies. Specifically, the objectives of this review article are: (i) to provide an overview of groundwater quality monitoring; (ii) to analyze the detection, monitoring, and mitigation of contaminants in groundwater; (iii) to assess existing mitigation strategies in groundwater contaminants; (iv) to identify research gaps; and (v) to recommend best practices for groundwater monitoring and mitigation.

Unlike global review articles that broadly examine technologies or hydrogeological trends, this paper offers a context-specific synthesis focused on the Philippines. This review is important as it not only compiles global monitoring technologies and remediation strategies but also contextualizes them within the distinct urban and institutional challenges of heavily urbanized areas in the Philippines, providing actionable insights for researchers, practitioners, and policymakers.

2. Urbanization and Groundwater Challenges in the Philippines

The Philippines is undergoing rapid urban growth, with more people moving into cities in search of better opportunities and services [24]. As of 2020, nearly half of the population (59 million) lives in urban areas, a number that is expected to continue to rise.

Table 1 presents the total population, urban population, and corresponding urbanization levels for the Philippines and its regions for the years 2015 and 2020 [24]. Nationally, the proportion of the population residing in urban areas increased modestly from 51.2% in 2015 to 54.0% in 2020, reflecting ongoing urban growth driven by internal migration, economic concentration, and urban expansion. The National Capital Region (NCR) consistently remains fully urbanized, housing over 13 million people entirely within urban settings. In contrast, other regions show wide variation in urbanization levels, indicating differing degrees of urban development and infrastructure capacity.

Regions adjoining the NCR, notably Region III (Central Luzon) and Region IV-A (CALABARZON), exhibit relatively high urbanization rates of 66.3% and 70.5%, respectively, in 2020, underscoring their role as peri-urban spillover zones that absorb population and economic activities from Metro Manila. Likewise, Region XI (Davao Region) and Region XII (SOCCSKSARGEN) have also experienced notable increases in urban share, reflecting rapid growth of regional urban centers such as Davao City and General Santos City.

On the other hand, regions like Eastern Visayas (14.7%), Cagayan Valley (19.5%), and the Bicol Region (23.8%) maintain predominantly rural populations, with relatively slow urbanization progress over the 5-year period. Such disparities highlight the uneven spatial distribution of urbanization across the archipelago, which has implications for equitable resource allocation, infrastructure development, and provision of basic services.

The observed urbanization trends imply intensified pressures on water supply systems, sanitation networks, and particularly on groundwater resources in rapidly urbanizing regions since many of these areas do not have a reliable surface water source.

Figure 1 shows the groundwater availability map of the Philippines, sourced from the publicly accessible national geoportal (geoportal.gov.ph). The highlighted sections focus on key urban centers: Metro Manila, Cebu, and the Davao Region. It can be observed that Metro Manila and the Davao Region possess relatively substantial groundwater reserves, providing important sources for domestic, commercial, and industrial uses. In contrast, Cebu exhibits comparatively limited groundwater availability, which underscores the region's greater vulnerability to water scarcity and the risk of over-extraction. This spatial variability in groundwater reserves emphasizes the need for region-specific management strategies to ensure sustainable utilization, particularly in rapidly urbanizing areas. Only the groundwater availability map is available publicly, and access for other maps, like type of industries, geological, and geographical context, is not available, which is considered a limitation in the analysis.

Table 1. Population and urbanization increase in the Philippines.

Region, Province, and Highly Urbanized City	Total Population		Urban Population		Percent Urban	
	2020	2015	2020	2015	2020	2015
PHILIPPINES	109,033,245	100,979,303	58,930,729	51,728,697	54.0	51.2
NATIONAL CAPITAL REGION (NCR)	13,484,462	12,877,253	13,484,462	12,877,253	100.0	100.0
CORDILLERA ADMINISTRATIVE REGION (CAR)	1,797,660	1,722,006	598,688	524,672	33.3	30.5
REGION I (ILOCOS REGION)	5,301,139	5,026,128	1,351,205	1,029,562	25.5	20.5
REGION II (CAGAYAN VALLEY)	3,685,744	3,451,410	717,788	663,695	19.5	19.2
REGION III (CENTRAL LUZON)	12,422,172	11,218,177	8,230,254	6,914,703	66.3	61.6
REGION IV-A (CALABARZON)	16,195,042	14,414,774	11,415,742	9,564,515	70.5	66.4
MIMAROPA REGION	3,228,558	2,963,360	1,138,021	905,666	35.2	30.6
REGION V (BICOL REGION)	6,082,165	5,796,989	1,447,370	1,344,903	23.8	23.2
REGION VI (WESTERN VISAYAS)	7,954,723	7,536,383	3,353,205	2,868,795	42.2	38.1
REGION VII (CENTRAL VISAYAS)	8,081,988	7,396,898	4,196,639	3,656,628	51.9	49.4
REGION VIII (EASTERN VISAYAS)	4,547,150	4,440,150	666,473	529,902	14.7	11.9
REGION IX (ZAMBOANGA PENINSULA)	3,875,576	3,629,783	1,489,443	1,373,274	38.4	37.8
REGION X (NORTHERN MINDANAO)	5,022,768	4,689,302	2,528,239	2,272,001	50.3	48.5
REGION XI (DAVAO REGION)	5,243,536	4,893,318	3,504,533	3,108,872	66.8	63.5
REGION XII (SOCCSKSARGEN)	4,360,974	4,053,514	2,418,843	2,031,361	55.5	50.1
REGION XIII (CARAGA)	2,804,788	2,596,709	1,027,223	869,195	36.6	33.5
BARMM	4,944,800	4,273,149	1,362,601	1,193,700	27.6	27.9

Presented in Table 2 is the country's generally increasing trend in total water withdrawals over the past decade, rising from approximately 83.34 billion cubic meters (bcm) in 2010 to over 91 bcm by 2022 [25]. This gradual growth in total extraction reflects the country's expanding demand for water driven by economic development, agricultural intensification, and rapid urban population growth. Despite this increase in the absolute volume of water withdrawn, the national level of water stress has also continued to rise—from about 25.5% in 2010 to nearly 28% in 2022.

This pattern indicates that additional extraction alone has not sufficiently met the mounting demand pressures, especially in the context of a steadily growing population, urban expansion, and increased per capita consumption. The persistent and slightly rising water stress level suggests that water resources are becoming more strained relative to available supply, highlighting systemic inefficiencies in water allocation, limitations in storage and distribution infrastructure, and the degradation of water quality in many catchments and aquifers.

If these trends persist without significant improvements in water-use efficiency, integrated resource management, and demand-side controls, the country may face heightened risks of local water scarcity, particularly during dry seasons and extreme climate events. Therefore, managing both extraction volumes and overall demand, alongside

protecting source water quality, is essential to lowering national water stress and ensuring long-term water security.

The information provided in the map at Figure 1 focused on



Figure 1. Groundwater availability in the Philippines (Source: Geoportal ph).

Table 2. Total withdrawals and level of water stress in the Philippines.

Year	Total Water Withdrawals (in mcm)	Total Water Withdrawals (in bcm)	Level of Water Stress
2010	83,336.25	83.34	25.48%
2011	84,104.91	84.10	25.71%
2012	84,292.17	84.29	25.77%
2013	84,482.09	84.48	25.83%
2014	84,628.06	84.63	25.87%
2015	85,556.47	85.56	26.16%
2016	86,298.74	86.30	26.38%
2017	91,920.20	91.92	28.10%
2018	92,282.29	92.28	28.21%
2019	85,994.51	85.99	26.29%
2020	87,477.27	87.48	26.74%
2021	89,000.36	89.00	27.21%
2022	91,036.89	91.04	27.83%

Over-extraction of groundwater has emerged as a critical issue in many countries [26], including the Philippines, particularly its urban centers due to the escalating demand

for water driven by population growth, industrialization, and inadequate surface water supplies. Metro Manila, Cebu City, and Davao City are among the urban areas where excessive withdrawal has led to significant declines in groundwater levels. This unsustainable abstraction not only depletes aquifers faster than their natural recharge rates but also results in adverse consequences such as land subsidence, reduced water quality, and increased risk of saltwater intrusion in coastal cities [27]. Land subsidence linked to overpumping has already been documented in parts of Metro Manila, exacerbating the risk of flooding during heavy rainfall and high tides [28]. Moreover, lowering the water table can concentrate contaminants and make shallow wells more vulnerable to pollution from leaking septic tanks and urban runoff.

However, urbanization has brought a range of challenges to groundwater quality and availability. The fast pace of development, often unregulated, has led to land-use changes and the expansion of residential and industrial zones without proper environmental planning [29]. Informal settlements, which often rely on makeshift septic systems, are common in densely populated areas. These systems can leak and contaminate nearby groundwater sources with pathogens like E. coli and total coliforms, raising serious public health concerns [30]. Even with the availability of supplied water in some urban areas, some communities still rely on groundwater for drinking and domestic use, which basically poses serious public health risks due to persistent contamination and inadequate urban sanitation infrastructure. Densely populated cities such as Metro Manila, Metro Cebu, and Davao partially rely on deep and shallow wells, yet many urban households and informal settlements use poorly constructed septic tanks or open defecation areas that leak pathogens directly into the subsurface. Studies have shown that urban groundwater in Metro Manila frequently contains *Escherichia coli* and total coliforms at levels exceeding safe limits, heightening the incidence of diarrheal diseases, gastroenteritis, and other waterborne infections [31,32]. Unlike developed nations with advanced centralized water treatment infrastructure and well-enforced standards, many areas in the country still lack sufficient facilities to effectively remove microbial and chemical pollutants from extracted groundwater before it is distributed for domestic use [33].

Industrial activities and improper waste disposal further contribute to the problem. In areas like Cebu and Pampanga, studies have found elevated levels of heavy metals such as arsenic, lead, and cadmium in groundwater, particularly near landfills and former mining sites [13,34]. These contaminants often exceed safe limits for drinking water and can cause long-term health problems. Urbanization also affects the natural recharge of groundwater. As more land is covered by concrete and asphalt, less rainwater is able to seep into the ground and replenish aquifers. At the same time, increased surface runoff during heavy rains can carry pollutants into shallow groundwater systems, especially in areas with porous soil or fractured rock formations [35].

In recent years, emerging contaminants like pharmaceuticals, personal care products, and volatile organic compounds (VOCs) have been detected in urban environments [36], particularly in Southeast Asian regions like the Philippines [37]. These substances enter the water system through untreated wastewater and landfill leachate and are not yet fully addressed in current regulations. Their long-term effects on ecosystems and human health are still being studied, but their presence highlights the need for more comprehensive monitoring.

Compounding these challenges are emerging threats from climate change. Rising sea levels accelerate saltwater intrusion into coastal aquifers, a growing issue in Metro Manila's reclaimed areas and coastal Cebu [38]. Intense rainfall events increase runoff, washing urban pollutants into shallow groundwater layers through permeable soils and unsealed drainage channels.

Managing these challenges is complicated by the country's fragmented water governance. Groundwater management responsibilities are shared among national agencies like the Department of Environment and Natural Resources (DENR) and the National Water Resources Board (NWRB), as well as local government units. Unfortunately, coordination between these agencies is often limited, leading to gaps in monitoring, data sharing, and enforcement. Also, improving this situation requires investing in modern treatment technologies, strengthening regulatory compliance, expanding community-level water safety planning, and fostering public awareness on water safety. Without these, the continued dependence on untreated or poorly treated groundwater poses a persistent threat to urban public health and sustainable urban development in the country.

Overall, the combination of rapid urban expansion, outdated infrastructure, and limited institutional capacity makes groundwater management in the Philippines particularly difficult. Addressing these issues will require not only better technology and monitoring systems but also stronger policies, better coordination between agencies, and greater public awareness about protecting this vital resource.

To address these multifaceted issues, there is a pressing need to modernize urban water infrastructure through investment in robust treatment systems, improved sewerage networks, and the rehabilitation of existing wells and distribution pipelines. Equally important are policy reforms to strengthen regulatory oversight, enforce groundwater extraction limits, and promote equitable access to safe drinking water. Enhanced interagency coordination and data integration would allow for more accurate resource assessments and targeted interventions. Also, raising community awareness about the risks of groundwater overuse and contamination, coupled with participatory water governance, can foster more sustainable practices at both household and community levels. Together, these measures are essential to ensure that urban groundwater remains a safe, resilient, and sustainable resource in the face of continuing urbanization and climate pressures.

3. Review Methodology

This review adopts a meta-analysis approach to evaluate groundwater assessment and contaminant monitoring strategies in heavily urbanized environments, with specific attention to their application and relevance in the Philippine context. The analysis draws from a comprehensive pool of 162 references, including peer-reviewed journal articles, institutional reports, technical papers, and web-based sources.

The included studies were synthesized to identify prevailing monitoring practices, commonly assessed contaminants, and geographic trends. Special attention was provided to identifying underexplored contaminant categories, emerging technologies lacking local validation, and systemic implementation gaps. The evidence was further structured using visual tools such as evidence maps, conceptual frameworks, and temporal distribution charts. These analyses informed the formulation of key findings, limitations, and future directions discussed in the succeeding sections.

3.1. Literature Sourced and Selection Criteria

The literature search was designed to capture a diverse yet targeted selection of studies that address the state of groundwater contamination and monitoring innovations. The Scopus database served as the primary source for peer-reviewed articles due to its wide coverage of environmental and engineering literature. Supplementary references were gathered from institutional repositories such as the World Water Quality Alliance (WWQA) and the International Association of Hydrogeologists (IAH), as well as reliable online platforms, government publications, and international conference proceedings.

The search was limited to publications from 2000 to 2025, allowing the inclusion of both foundational work and the most recent technological developments. Studies were

screened based on their relevance to groundwater quality in urban or peri-urban settings, with an emphasis on those that offered empirical evidence, introduced innovative monitoring tools, or discussed integrated management and planning approaches.

To organize the review coherently, the references were initially grouped into two overarching thematic lenses: (1) monitoring technologies, which include applications of the Internet of Things (IoT), real-time sensor systems, remote sensing and GIS, artificial intelligence (AI), machine learning (ML), biosensors, and DNA-based microbial detection; and (2) contaminants, which include heavy metals, nitrate and pesticide runoff, volatile organic compounds (VOCs), chlorinated solvents, hydrocarbons, pathogens, and pharmaceutical and personal care products (PPCPs).

This classification allowed for a focused synthesis of methods, findings, and contextual applications across different pollution scenarios. It should be noted that the referenced studies employed various extraction and analytical methods for contaminant detection. As such, cross-comparisons and synthesized conclusions may involve small to substantial margins of error due to methodological variability.

3.2. Thematic Classification

To enable a structured synthesis of the reviewed literature, all 161 references were categorized into five overlapping thematic groups. These categories reflect the major research domains addressed by the selected studies and provide a comprehensive lens through which to interpret trends, gaps, and opportunities in groundwater contamination and monitoring research. The classification is not mutually exclusive, as several references contribute to multiple thematic areas.

The first group, groundwater contamination and pollutant profiles, includes studies that investigate the sources, distribution, and risk impacts of specific pollutants in urban and peri-urban groundwater systems. These works address contaminants such as arsenic, mercury, nitrate, pesticides, hydrocarbons, pathogens, and VOCs. They form the scientific basis for understanding the extent and variability of contamination, especially in relation to land use and anthropogenic activity.

The second group, monitoring technologies and assessment methods, represents the largest category. These studies explore the development and application of modern tools, such as wireless sensor networks, remote sensing and Geographic Information System (GIS), artificial intelligence (AI), machine learning (ML), biosensors, and DNA-based detection systems. Many of these works emphasize real-time or predictive monitoring capabilities, enabling early detection and data-driven responses to groundwater threats.

The third thematic group, remediation techniques and planning integration, includes literature focused on mitigating contamination and incorporating groundwater risk into land-use decision-making. These studies discuss methods such as permeable reactive barriers (PRBs), bioremediation, advanced membrane filtration, and the use of vulnerability mapping tools like DRASTIC, which stands for D—depth to groundwater, R—recharge rate, A—aquifer, S—soil, T—topography, I—vadose zone's impact, and C—aquifer's hydraulic conductivity. Several papers also highlight the need for harmonizing environmental monitoring with zoning policies and infrastructure planning.

The fourth group, Philippine-specific case studies and governance insights, includes empirical studies, community-based assessments, and policy reviews specific to the Philippines. These references offer valuable insights into site-specific contamination patterns, the limitations of local monitoring frameworks, and the challenges of institutional coordination. Examples are drawn from Metro Manila, Cebu, Tarlac, Pampanga, and other regions.

Finally, a fifth category, reviews and global frameworks , comprises broad-scope literature that provides conceptual models, synthesis papers, and global perspectives on

groundwater contamination and sustainable management. These include reports from the World Water Quality Alliance, Sustainable Development Goals (SDG)-related publications, and methodological overviews that helped shape the structure of this review.

The classification is summarized in Table 3.

Table 3. Summary of thematic classification of literature cited in this study.

Focus Area	Approximate No. of References
Groundwater contamination and pollutant profiles	58
Monitoring technologies and assessment methods	56
Remediation techniques and planning integration	35
Philippine-specific case studies and governance	42
Reviews and global frameworks (cross-cutting)	34

Note**:** Some references appear in multiple categories. The counts reflect thematic groupings rather than mutually exclusive sets and are rounded for clarity.

3.3. Temporal and Source Distribution

The publication years of the 161 references were analyzed to assess the timeliness and evolution of research trends. The references were grouped into five categories based on their year of publication, as shown in Figure 2. The results show that more than half of the references were published in the past 7 years (2018–2025), emphasizing that the review is grounded in recent innovations and updated understandings of groundwater monitoring and contaminants. The inclusion of earlier works ensures continuity with foundational knowledge and established methodologies.

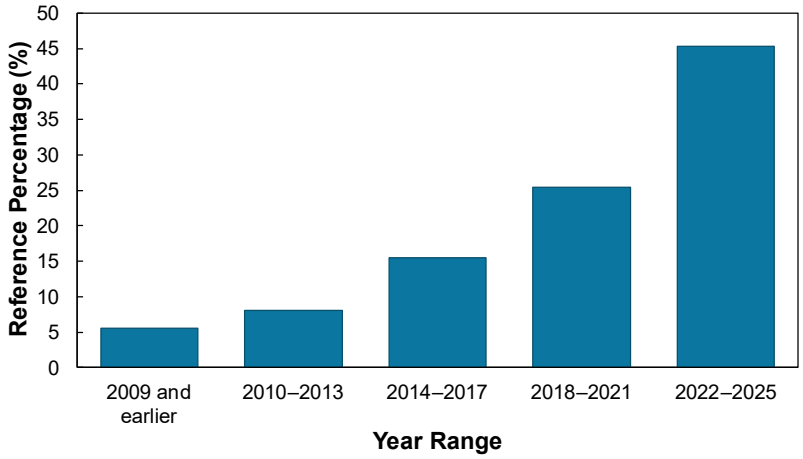


Figure 2. Reference distribution based on year of publication.

Moreover, out of the 162 references reviewed in this study, 21% of the publications were identified as Philippine-specific, providing localized insights into groundwater contamination, monitoring practices, and governance issues across regions such as Metro Manila, Cebu, Tarlac, and Pampanga. These studies offer crucial empirical data and contextual understanding that inform the relevance of global technologies to the Philippine setting. The remaining references represent global or international studies that contribute a broader view of technological advancements, contaminant trends, and best practices in groundwater quality assessment. This distribution reflects a strong reliance on international literature while highlighting the need for increased local research to strengthen context-sensitive policy and monitoring frameworks in the Philippines.

4. Groundwater Monitoring Methods and Technique

This section reviews the primary methods for assessing groundwater quality in urbanized settings, focusing on technological integration, artificial intelligence, and emerging innovations that enhance detection accuracy, timeliness, and remediation potential.

4.1. Technology Integration in Groundwater Monitoring

Technologies such as wireless sensor networks (WSNs), GIS-based mapping tools, and the DRASTIC vulnerability model have long been established in environmental science. However, their recent applications in real-time, AI-driven, and integrated systems—particularly for urban groundwater monitoring in developing nations—represent a significant shift from traditional implementations. This review emphasizes the emerging adaptations of these established tools in the context of data-scarce, rapidly urbanizing environments like the Philippines. Rather than focusing on technological novelty, the goal is to examine how mature technologies are being modernized and locally adapted to meet complex groundwater monitoring needs.

4.1.1. IoT-Based Monitoring Systems and Remote Sensing

The integration of Internet of Things (IoT) technologies in groundwater monitoring has enabled real-time data acquisition from multiple parameters such as pH, salinity, turbidity, and electrical conductivity [39]. Wireless Sensor Networks (WSNs), paired with microcontrollers and cloud-based platforms, are increasingly used to transmit groundwater quality data in real time. For instance, Oppus et al. (2020) [40] developed a sensor network in the Philippines that achieved 96.63% accuracy and allowed cloud-based visualization of pH, Electric Conductivity (EC), Total Dissolved Solids (TDS), and salinity. This system not only enhanced monitoring reliability in remote areas but also enabled automated alerts to prevent groundwater misuse. Similar systems have also been tested in Korea and India, with successful integration into smart water infrastructure. Such developments underscore the practicality of IoT-based monitoring, especially in countries with limited access to laboratory infrastructure [41,42]. These advancements in technology are crucial for improving groundwater quality monitoring, particularly in regions facing challenges related to resource limitations and environmental pressures.

Recent literature illustrates how these mature technologies are actively evolving. For example, Essamlali et al. (2024) reviewed real-time water quality monitoring systems that combine IoT sensors with machine learning for predictive analysis of key groundwater parameters [43]. These advancements enable early-warning capabilities and improve data-driven decision-making in water-scarce urban settings. As the review progresses, other spatial and model-based tools, including GIS-integrated DRASTIC, are examined in terms of their modern adaptations and relevance.

4.1.2. Remote Sensing and GIS

Remote sensing techniques, supported by Geographic Information Systems (GIS), are also instrumental in detecting groundwater contamination patterns. Studies such as Yoon et al. (2024) [44] demonstrate how EC, Cl, NO_3-N, and NH_4-N parameters can be effectively monitored through remote sensors, especially in areas impacted by livestock waste and leachate. GIS mapping facilitates a spatial understanding of pollution hotspots, allowing for targeted investigation and resource allocation. In urban environments, this integration has proven useful for planning water safety interventions and assessing environmental risks [45–47].

Geographic Information Systems (GIS) continue to play a pivotal role in groundwater monitoring and vulnerability assessment, offering advanced capabilities for spatial

data analysis, integration, and visualization. Recent developments have enabled GIS platforms to interface with real-time sensor networks, land-use change datasets, and contamination detection models, thus supporting dynamic groundwater risk assessments [48,49].

For example, modern GIS applications are now being used to incorporate population density, road infrastructure, and impervious surface data to better understand the human-induced pressures on aquifers in urban areas [50]. Moreover, GIS is increasingly integrated with AI-driven modeling to enhance spatial resolution, predictive accuracy, and decision-support mapping, especially in complex urban environments. These innovations strengthen GIS's role not only as a mapping tool but as a decision-making framework for proactive groundwater management [51].

4.1.3. DRASTIC Method

The DRASTIC method (Depth to water, Recharge, Aquifer media, Soil media, Topography, Impact of vadose zone, Conductivity of aquifer) provides a vulnerability assessment model that identifies zones at higher risk of groundwater contamination [52–55]. It combines multiple hydrogeological layers using a weighted index to generate a vulnerability map. The method has been widely applied for land-use planning and policy formulation, such as in aquifer protection zones and site selection for waste disposal [56]. While effective for identifying high-risk areas, DRASTIC requires extensive datasets and does not provide real-time updates, making it more suitable for baseline assessments than continuous monitoring.

While the DRASTIC model remains a classic approach for groundwater vulnerability mapping, its underlying methodology has undergone significant refinement in recent years. Traditional DRASTIC uses fixed weighting for seven environmental parameters, but newer studies now apply entropy-based weighting, analytic hierarchy processes (AHP), or hybrid statistical learning techniques to improve objectivity and reduce bias in vulnerability scoring [57,58].

For instance, a recent study implemented an entropy-weighted DRASTIC method in an urban aquifer setting, resulting in improved spatial resolution and site-specific risk differentiation [59]. Similarly, recent studies demonstrated that integrating DRASTIC with multi-criteria decision-making (MCDM) techniques enhances the model's adaptability to varied hydrogeological conditions. These updated approaches reflect how DRASTIC remains highly relevant, particularly when recalibrated for use in data-scarce, urbanized environments such as those found in urban cities [60,61].

4.2. AI and Real-Time Data Monitoring

This approach leverages real-time data systems and intelligent algorithms to provide continuous tracking and analysis, enabling early detection of contaminants and better water resource management

4.2.1. Artificial Intelligence and Machine Learning

Machine-learning (ML) and artificial intelligence (AI) models are increasingly used to predict groundwater quality indices (WQI) and optimize monitoring networks [62,63]. For instance, Al-Adhaileh et al. (2022) [64] implemented Bidirectional Long Short-Term Memory (BiLSTM) and Adaptive Neuro-Fuzzy Inference System (ANFIS) models, which demonstrated superior accuracy over traditional regression approaches. These models learn from historical and sensor data to forecast contamination trends, making them useful for preemptive decision-making. Other models, such as Artificial Neural Network (ANN) and Support Vector Machine (SVM), have also been widely used in hydrology for parameter optimization and anomaly detection [65,66]. However, their accuracy depends heavily on the availability of well-calibrated input data. These advancements in AI and

machine learning enhance the predictive capabilities of groundwater monitoring systems, ultimately contributing to more effective management of water resources and contamination risks [65,66].

4.2.2. Real-Time Data Analytics

Real-time analytics enhances the utility of AI by enabling continuous learning from live data. When linked with IoT devices, AI systems can generate early warnings for sudden shifts in water quality, such as pH spikes or increased salinity due to leachate intrusion [63,67–69]. These systems offer timely intervention capabilities, which are critical in densely populated urban areas. However, challenges such as sensor drift, data loss, and signal interference can limit their performance. Calibration and redundancy protocols are necessary to maintain reliability [63,64]. Moreover, the integration of AI with IoT technologies not only enhances monitoring precision but also facilitates proactive management strategies that can mitigate contamination risks effectively [70].

Although AI and machine-learning models are well-established in hydrology, recent implementations have focused on real-time integration with IoT systems, anomaly detection, and hybrid decision-support tools for urban groundwater risk management [71,72]. These advancements demonstrate that the focus is shifting from theoretical modeling to practical, context-aware prediction systems.

4.2.3. Limitations and Challenges

While artificial intelligence (AI) and machine learning (ML) hold significant promise for advancing groundwater contamination monitoring and predictive modeling, several critical limitations constrain their practical effectiveness, particularly in data-scarce contexts such as the Philippines. One of the foremost challenges is the limited availability of high-resolution, long-term, and site-specific hydrochemical and hydrogeological datasets required to train ML models. Sparse and inconsistently collected data increase the risk of overfitting, where models may perform well on training data but fail to generalize to new or changing conditions. Moreover, the absence of adequate local calibration data undermines the reliability of predictive outputs, as many existing models are adapted from data-rich regions with different geological, climatic, and anthropogenic contexts. Another notable issue is sampling bias, often resulting from irregular monitoring, varying laboratory standards, and inconsistent spatial coverage, which can introduce systematic errors into training datasets. These constraints highlight that, without substantial investments in improving monitoring infrastructure and promoting open-access data sharing, the full potential of AI/ML approaches in groundwater management cannot be realized. Thus, integrating AI/ML must be approached cautiously and complemented with robust field investigations and expert validation to ensure scientifically defensible and locally relevant outcomes.

4.3. Emerging Groundwater Monitoring Techniques

This includes cutting-edge methods such as biosensors and DNA-based methods, which are designed to enhance the precision, coverage, and affordability of groundwater monitoring efforts.

4.3.1. Biosensors

Biosensors offer an alternative to traditional methods, which can be time-consuming, expensive, and labor-intensive [73], providing an in-depth analysis of the application of biosensors in wastewater monitoring. Biosensors are analytical devices that combine a biological recognition element (e.g., enzymes, microbes, or antibodies) with a physicochemical detector [74–76]. They can detect specific pollutants and contaminants, making them

suitable for monitoring various environmental pollutants in wastewater. The research highlights the potential of biosensors as innovative tools for improving wastewater monitoring, offering a faster, more efficient, and cost-effective alternative to conventional methods. The review calls for continued advancements in biosensor technology to address existing limitations and expand their practical applications in environmental management [73,77]. These innovative techniques are essential for enhancing groundwater monitoring, particularly in urbanized regions where traditional methods may be inadequate or impractical.

Although biosensors have been explored for decades, recent innovations—particularly those involving nanomaterials and portable sensing platforms—have made them viable for field deployment. Recent studies describe how nano-biosensors are being used for on-site monitoring of heavy metals and microbial contaminants with improved sensitivity and real-time capability [78,79].

4.3.2. DNA-Based Methods

The sequencing of rRNA or DNA is used by microbial communities as an indicator of water quality and sources of contamination [80–82]. Lyons et al. (2022) [83] explore innovative approaches to groundwater quality monitoring, combining real-time data collection, stable water isotopes, and microbial community analysis. Combining real-time data collection with stable water isotopes and microbial community analysis, the findings demonstrated that real-time data collection methods worked well with traditional methods and added new material. This suggests the advanced monitoring techniques being developed by the group will provide much new information in groundwater quality monitoring and management over the years to come [84].

4.4. Summary of Groundwater Monitoring Techniques

Table 4 summarizes the groundwater monitoring methods discussed, highlighting their description, key advantages, and common limitations. The summarized techniques highlight the breadth of current capabilities in groundwater monitoring across technological, analytical, and biological domains. Technology-based systems such as IoT monitoring and GIS-integrated remote sensing are highly scalable and beneficial for continuous surveillance, especially in data-scarce regions [40,44]. However, they require upfront investments in infrastructure and data transmission reliability. AI and machine-learning applications offer predictive power and cost efficiency for long-term groundwater quality management, especially when integrated with real-time data feeds [62–64]. Despite their promise, they are dependent on large, high-quality datasets and technical expertise, which may limit widespread adoption in developing settings.

Table 4. Comparison of groundwater monitoring techniques: technologies, applications, and key features.

Category	Technique	Description	Advantages	Challenges	Applicable Contaminants	References
Technology-Based	IoT-Based Monitoring	Real-time data acquisition using sensor networks and microcontrollers	Continuous monitoring, cost-effective, scalable	Initial cost, sensor maintenance, data transmission issues	pH, EC, salinity, turbidity, nitrates	[39–42]
	Remote Sensing & GIS	Remote sensors and GIS tools for spatial analysis of contaminants	Large-scale coverage, mapping capabilities	Affected by weather, resolution limitations	Nitrate, chloride, ammonium	[44–47]

	DRASTIC Method	Hydrogeological vulnerability mapping	Identifies high-risk zones, widely accepted	Data-intensive, lacks temporal dynamics	Multiple pollutants (risk zones only)	[52–56]
AI & Real-Time	Machine Learning (ANN, SVM, ANFIS)	Predictive models for groundwater quality indices	High accuracy, reduces lab analysis needs	Requires large datasets and training expertise	Heavy metals, nitrates, multi-parameter indices	[62–66,85,86]
	Real-Time Analytics	Integration of AI with real-time sensor data	Early warning, fast response	Dependent on sensor reliability and network access	Sudden shifts in water quality (multi-contaminant)	[63,64,67–71]
Emerging	Biosensors	Analytical tools with biological recognition for specific pollutants	Fast, sensitive, low-cost, on-site detection	Limited target range, lifespan and calibration issues	Heavy metals, PPCPs, pathogen	[73–77]
	DNA-Based Methods	Microbial fingerprinting using sequencing data	Identifies contamination source and ecology	Costly, complex interpretation	Pathogens (E. coli, etc.)	[80–84]

Emerging tools like biosensors and DNA-based methods push the frontier of specificity and sensitivity. These methods are particularly useful for targeted monitoring of pathogens and heavy metals, offering fast and localized results [73]. Nonetheless, their use is currently limited by technical complexity and cost barriers.

Taken together, these techniques are not mutually exclusive. An integrated approach—leveraging real-time sensors, predictive modeling, and targeted biosensing—offers the most promising pathway for effective groundwater quality assessment in urbanized environments, especially in places like the Philippines where infrastructure and contaminant diversity vary widely.

5. Groundwater Contaminants Due to Heavy Urbanization

Urbanization significantly alters natural hydrological cycles, leading to increased impervious surfaces, reduced recharge, and heightened pollutant loads entering groundwater systems [87,88]. In highly urbanized areas of the Philippines, groundwater remains a crucial water source, making the assessment and management of urban-derived contaminants imperative [40]. This section summarizes key contaminant groups, their sources, effects, detection techniques, and remediation strategies.

5.1. Common Groundwater Contaminants

The data in Table 5 highlight the variety and severity of groundwater contaminants observed in urban Philippine environments. Heavy metals like lead, cadmium, and arsenic are persistent pollutants commonly associated with landfills and industrial activities; for instance, elevated levels of lead have been recorded near the Payatas landfill [89]. These metals pose severe health risks, including neurological damage and organ failure [90]. Furthermore, the presence of these heavy metals in urban groundwater can be attributed to industrial discharge and improper waste management practices, necessitating urgent remediation efforts [91,92].

Table 5. Summary of common urban groundwater contaminants, sources, and risk.

Contaminant Group	Examples	Primary Sources	Reported Levels	Environmental/ Health Risks	References
Heavy Metals	Lead, Arsenic, Cadmium, Chromium	Industrial discharge, mining, landfills	Lead—up to 0.03 mg/L (Payatas); Arsenic—0.01–0.05 mg/L (Pampanga)	Neurotoxicity, carcinogenicity, kidney damage	[62–66]

Nutrients & Pesticides	Nitrate, Ammonia, Glyphosate	Agricultural runoff, septic leakage	Nitrate—up to 72 mg/L (Santa Ignacia, Tarlac)	Eutrophication, blue baby syndrome, hormone disruption	[82–84]
VOCs & Chlorinated Solvents	Benzene, TCE, PCE	Industrial solvents, fuel stations	Benzene—>5 μg/L in select industrial zones	Carcinogenicity, liver/kidney effects	[88–91]
Pathogens	*E. coli*, *Salmonella*	Improper sewage disposal, septic tank leakage	Total coliforms and E. coli present in La Union wells	Gastrointestinal infections, waterborne disease outbreaks	[92,93]
Hydrocarbons	Toluene, Xylene, Diesel	Oil spills, underground storage tanks	Presence reported near fuel depots (quantitative data limited)	Soil and water contamination, chronic health impacts	[89]
Pharmaceuticals & Personal Care Products (PPCPs)	Antibiotics, hormones	Hospital waste, sewage effluent	No quantitative Philippine data yet; global detection at ng/L to μg/L levels	Antibiotic resistance, endocrine disruption	[46]

Nutrient-related pollution, particularly nitrate contamination, has been consistently linked to agricultural practices and poor waste management infrastructure, particularly in peri-urban areas [93–95]. In regions with high fertilizer use and unregulated septic systems, nitrate levels frequently surpass WHO-recommended thresholds, posing risks such as methemoglobinemia or "blue baby syndrome" [96–99].

Volatile organic compounds (VOCs) and chlorinated solvents, such as benzene and TCE, are more insidious due to their mobility in groundwater and strong links to liver and kidney damage. These contaminants are often found in groundwater near fuel stations and industrial zones [100–103].

Pathogenic contaminants like E. coli and Salmonella, though biological in nature, remain a public health concern in areas with poor sanitation infrastructure. These are prevalent in groundwater samples from densely populated informal settlements and areas with aging or failed septic systems [104–107].

Hydrocarbons such as diesel, toluene, and xylene are frequent byproducts of petroleum leaks and spills. Their persistence in aquifers and bioaccumulative properties threaten both ecological and human health [101,108–111].

Pharmaceuticals and personal care products (PPCPs) are increasingly detected in groundwater due to insufficient treatment of hospital and domestic waste. Compounds such as sulfamethoxazole and synthetic hormones persist in the environment and can induce microbial resistance and endocrine disruption [73,95,112,113].

These findings underscore the need for multi-contaminant detection systems and integrated management approaches tailored to local urban contexts. The diverse range of contaminants found in urban groundwater highlights the urgent need for effective monitoring and remediation strategies to safeguard public health and environmental integrity. Implementing comprehensive groundwater management strategies is essential to address the complex interplay of contaminants and ensure safe drinking water for urban populations. According to the comprehensive review by Lapworth et al. (2012) [114], the pervasive contamination of groundwater with a broad spectrum of emerging contaminants, including pharmaceuticals, personal care products, endocrine-disrupting compounds, and industrial chemicals, results from both contemporary anthropogenic inputs and historical land-use practices. These contaminants are increasingly detected at concentrations that may pose ecotoxicological risks and potential human health concerns. Without heightened public awareness and effective community engagement, daily activities that contribute to the introduction and persistence of these substances are likely to continue unchecked. This situation threatens not only the ecological integrity of groundwater-

dependent ecosystems but also compromises the safety and sustainability of potable groundwater supplies, particularly in densely populated urban centers.

5.2. Detection and Monitoring Strategies

Detecting urban-derived groundwater contaminants demands a hybrid approach that leverages both laboratory precision and field-deployable technologies [115–117]. For heavy metals, methods such as Atomic Absorption Spectrometry (AAS) and Inductively Coupled Plasma Mass Spectrometry (ICP-MS) are still the gold standards due to their accuracy in detecting trace metals like cadmium and arsenic [118,119]. However, their dependence on centralized labs and sample transport delays makes them less suitable for large-scale, real-time monitoring. In response, portable X-ray fluorescence (pXRF) analyzers and electrochemical biosensors have gained popularity for field use, especially in low-resource settings [120,121].

For nutrient contaminants such as nitrate and ammonia, standard detection methods include ion chromatography and UV-spectrophotometry [122,123]. Recent innovations include hybrid bio-inorganic sensors that simultaneously reduce and quantify nitrate levels, converting them into non-toxic forms while enabling continuous monitoring [97]. To understand regional patterns, spatial tools like Geographic Information Systems (GIS) and Inverse Distance Weighting (IDW) interpolation are employed to map nitrate plumes, revealing agricultural hotspots and vulnerability zones [98,124–126].

Volatile organic compounds (VOCs) and chlorinated solvents, including benzene and TCE, are commonly analyzed via gas chromatography–mass spectrometry (GC-MS) [127,128]. However, due to their volatility and subsurface transport behavior, in situ monitoring using fiber-optic sensors and multi-level piezometers is becoming increasingly important. Direct Push Technology (DPT) also allows rapid, stratified sampling to construct vertical contamination profiles [101,102].

For microbial pathogens such as E. coli, detection remains split between traditional culture-based enumeration methods and molecular techniques like qPCR (quantitative Polymerase Chain Reaction) [129–131]. Biosensors targeting microbial-specific enzymes or antigens are being developed for real-time pathogen detection, though they currently require careful calibration and often remain in the pilot-testing phase [104,105].

In monitoring PPCPs, LC-MS/MS (liquid chromatography–tandem mass spectrometry) offers superior resolution to identify and quantify low-concentration pharmaceuticals. Sensor-driven networks equipped with anomaly detection via AI/ML algorithms are under exploration to enhance early-warning systems, particularly in treatment plant inflow assessments [73,132,133].

Overall, integrating these techniques into a layered detection strategy—combining precision lab analysis with field sensors and spatial modeling—yields the most robust surveillance system, adaptable to varied urban contamination contexts.

5.3. Remediation Strategies

Remediating contaminated groundwater in urban areas presents unique challenges due to mixed pollution sources and subsurface complexities. The choice between in situ and ex situ methods is guided by factors such as aquifer depth, contaminant type, and the proximity of sensitive receptors [134,135].

In situ remediation methods have gained prominence due to their lower energy requirements and minimized environmental disturbance. Permeable Reactive Barriers (PRBs) are among the most studied in situ technologies. Installed perpendicular to groundwater flow, PRBs contain reactive media—such as zero-valent iron (ZVI), activated carbon, or zeolite—that trap or chemically transform contaminants [136,137]. PRBs are particularly effective for removing heavy metals, chlorinated solvents, and select

pathogens [138,139]. However, they require careful pre-installation site assessment to avoid clogging, ensure hydraulic connectivity, and match contaminant profiles.

Bioremediation techniques, which use native or introduced microorganisms to break down pollutants, offer a low-cost, environmentally friendly alternative for nitrate, hydrocarbons, and organic contaminants [140,141]. Approaches such as bioaugmentation (adding specific microbes) and biostimulation (adding nutrients to enhance native microbial activity) are applied based on site characteristics [139]. In anaerobic settings, sulfate- or nitrate-reducing bacteria can be used to remediate hydrocarbons and VOCs, though long treatment periods may be needed. Aerobic degradation, conversely, is faster but requires oxygen delivery systems.

Ex situ techniques such as pump-and-treat remain essential for acute contamination sites. Contaminated water is pumped to the surface, treated (e.g., using activated carbon, air stripping, or membrane filtration), and returned or safely discharged. While versatile, these systems are known for diminishing returns over time and high operational costs [142–144]. In recent years, integrated electrokinetic remediation technology has emerged as a promising in situ method for addressing subsurface contamination, offering both economic and environmental benefits [145]. This approach harnesses electrical fields to enhance the movement of contaminants toward recovery systems, thus improving remediation efficiency in heterogeneous environments.

Membrane-based technologies, particularly nanofiltration (NF) and reverse osmosis (RO), are widely used for treating PPCPs and VOCs. NF membranes remove larger organic compounds, while RO provides fine-level filtration [146]. However, they suffer from high energy requirements, membrane fouling, and concentrate disposal issues [103].

Recent advances focus on hybrid remediation, where technologies are combined to improve efficiency and address broader contaminant ranges. Examples include PRBs layered with bioactive media, coupled chemical reduction–electrocoagulation systems, and membrane–oxidation treatments [147–149]. These combinations increase contaminant removal rates while maintaining cost-efficiency [99].

Ultimately, selecting a remediation strategy requires a contaminant-specific, site-informed approach that considers hydrogeological conditions, regulatory thresholds, and long-term sustainability.

5.4. Integrated Perspective and Local Context

Before contextualizing the Philippine case, it is helpful to consolidate key insights from the previous sections. Table 6 summarizes common detection and remediation techniques aligned with specific contaminants prevalent in urban groundwater environments.

This table highlights how certain technologies, such as biosensors and PRBs, offer cross-contaminant utility, whereas others like RO and LC-MS/MS are tailored for complex organics and trace pollutants. Understanding these overlaps is critical for developing multi-contaminant monitoring frameworks, especially in resource-constrained urban settings.

Urban groundwater contamination in the Philippines is spatially diverse but thematically consistent—driven by overlapping industrial, agricultural, and domestic pressures [150]. In Metro Manila, nitrate and pathogen contamination is frequently detected near informal settlements with aging or absent septic infrastructure [82–84]. Simultaneously, heavy metals and hydrocarbons are prevalent near industrial zones, waste transfer stations, and fuel depots, with studies detecting lead, cadmium, and benzene in shallow aquifers across the metro [101,151].

The problem is compounded by seasonal patterns. During the rainy season, rapid infiltration of surface runoff accelerates contaminant migration into aquifers, particularly in areas lacking proper drainage or landfill lining. In cities like Cebu and Davao, karstic

geology exacerbates vertical contaminant flow, while poorly regulated landfills contribute to diffuse pollution plumes [152–155].

Table 6. Summary of detection and remediation techniques for major contaminants.

Contaminant Type	Common Detection Methods	Preferred Remediation Techniques	Remarks
Heavy Metals	ICP-MS, AAS, portable XRF [67,68,71]	PRBs with ZVI, chemical precipitation [74,80]	Requires site-specific media selection and long-term monitoring
Nitrate	Ion chromatography, UV spectrophotometry [83]	Bioremediation, denitrifying biofilters [74,81]	High mobility necessitates continuous tracking and multi-barrier systems
VOCs & Chlorinated Solvents	GC-MS, in situ fiber-optic sensors [89,90]	PRBs, chemical oxidation, air stripping [87,91]	Often found with hydrocarbons; require stratified sampling and modeling
Pathogens	qPCR, biosensors [92,93]	Bioremediation, chlorination [81]	Biosensor calibration is key for reliability in field deployment
Hydrocarbons	GC-MS, Direct Push Technology (DPT) [89]	Anaerobic bioremediation, phytoremediation [49,74]	Often co-occurs with metals and solvents; integrated approaches are ideal
PPCPs	LC-MS/MS, AI-integrated sensor networks [46]	Reverse osmosis, nanofiltration, hybrid oxidation-membrane [87,91]	Persistent in wastewater; not effectively removed by conventional systems

Despite existing monitoring protocols, local government units (LGUs) often face technical and financial limitations that hinder real-time assessment and long-term remediation [156,157]. Many municipalities still rely on laboratory-based sampling with limited spatial coverage and irregular intervals [16]. Integration of low-cost biosensors and mobile reporting platforms can substantially increase coverage and speed up community-level responses to contamination incidents.

Urban planning also plays a vital role. Many residential and industrial developments occur without incorporating groundwater vulnerability assessments. Tools such as GIS-based overlay analysis and machine-learning models can predict high-risk zones by combining land use, aquifer properties, and pollution history [27,158]. These models can guide zoning regulations, prioritize remediation funding, and shape future infrastructure development [27].

To close the governance gap, multisectoral coordination is essential. Environmental agencies, water utilities, LGUs, and academic institutions must co-develop frameworks for integrated monitoring, transparent reporting, and sustainable remediation. The development and enforcement of data-informed policies will be instrumental in reducing future groundwater contamination and ensuring safe water supply for urban populations [159].

6. Synthesis, Research Gaps, and Future Directions

6.1. Synthesis of Findings

The review in Sections 4 and 5 reveals the evolving landscape of groundwater contamination monitoring in urban areas, with a growing reliance on innovative technologies amid increasing environmental pressures. Section 4 focused on various monitoring techniques, such as IoT-based sensor networks, remote sensing, AI/ML predictive models, biosensors, and DNA-based methods. Section 5, on the other hand, examined key contaminants in urban groundwater systems—including heavy metals, nitrates, pathogens, hydrocarbons, VOCs, and PPCPs—and discussed their detection methods and remediation strategies.

Together, these sections demonstrate that while the scientific tools available for monitoring are becoming more sophisticated and capable, their application remains uneven

across contaminant types and geographic settings, particularly in the Philippines. Many of the contaminants discussed are prevalent in Philippine cities, but few studies integrate modern monitoring systems with local validation. For example, heavy metals have been reported in Cebu and Batangas, and nitrate and pathogens are a concern in areas like Tarlac and La Union. However, in most of these cases, conventional detection techniques prevail, with limited incorporation of AI, biosensors, or IoT technologies.

To visualize this synthesis, an evidence map was created (see Table 7), showing the overlap among contaminants, monitoring technologies, and their Philippine field application. This table is based on a combination of approaches: certain entries are directly supported by studies cited in Sections 4 and 5; others are inferred based on logical intersections between contaminant types and available monitoring technologies discussed in the literature. Additionally, some entries—particularly those referring to underexplored contaminants, such as PPCPs and VOCs—are introduced to reflect emerging global trends and to frame relevant research gaps in the Philippine context. This integrative approach was employed to enhance the comprehensiveness of the synthesis and support the conceptual framework and policy directions presented in later sections. The table shows that while AI and remote sensing are widely explored in the global literature, they remain underused in Philippine contexts. Moreover, contaminants like VOCs, hydrocarbons, and PPCPs are barely monitored locally despite their growing relevance in urban pollution.

Table 7. Evidence map of monitoring methods and groundwater contaminants in the Philippine context.

Contaminant Type	IoT/WSN	Remote Sensing/GIS	Biosensors	DNA-Based Methods	AI/ML	Field Application in PH	Research Gaps
Heavy Metals	✓	✓	✓	✘	✓	Cebu, Batangas, Pampanga	Integration of AI with real-time sensors
Nitrate and Pesticides	✓	✓	✘	✘	✓	Tarlac, La Union	Hybrid bio-inorganic sensors need local validation
VOCs and Chlorinated Solv.	✘	✘	✘	✘	✓	Few, not site-specific	Understudied, urgent risk
Pathogens (e.g., E. coli)	✓	✘	✓	✓	✘	La Union, Metro Manila	Biosensor field reliability
Hydrocarbons	✘	✘	✘	✘	✘	Rarely studied	Major knowledge gap
PPCPs (Pharma and Care Prods)	✘	✘	✘	✓	✓	None yet	No Philippine study so far

This synthesis reveals a fragmented research landscape, where technological advances exist but are not yet embedded in local practice. Therefore, any attempt to improve groundwater management must focus not just on adopting new technologies but on localizing and integrating them across monitoring, regulation, and remediation activities.

This review references several global frameworks, including the Sustainable Development Goals (e.g., SDG 6), the World Water Quality Alliance's groundwater quality assessments, and vulnerability mapping tools such as the DRASTIC model. While these frameworks offer robust guidance for integrated water management, their alignment with local practices in the Philippines is uneven. Institutional fragmentation, lack of real-time monitoring systems, and limited enforcement mechanisms often hinder the full adoption of global strategies. Bridging this gap requires localized adaptations of international tools and a stronger emphasis on policy integration and multi-stakeholder collaboration.

6.2. Conceptual Framework for Integrated Groundwater Monitoring

Based on the synthesis, a conceptual framework (Figure 3) was developed to illustrate how urban groundwater quality in the Philippines can be effectively assessed and managed. The framework connects the progression from pollution drivers to monitoring and remediation strategies, all within a context influenced by enabling and limiting factors.

The framework begins with urbanization-related drivers such as industrial activities, landfills, agricultural runoff, and septic tank leakage, which contribute to the degradation of groundwater quality. These drivers introduce a variety of contaminants, including heavy metals, nitrates, pathogens, and pharmaceuticals and personal care products (PPCPs), whose impacts vary in severity and persistence. To effectively monitor these risks, the framework includes a tiered set of technologies comprising real-time sensors, remote sensing platforms, GIS-integrated tools, AI/ML models, biosensors, and DNA-based microbial detection methods. These technologies generate data for decision-support systems such as vulnerability mapping, anomaly detection, and predictive risk modeling. Based on the monitoring outcomes, contaminant-specific remediation and management strategies are implemented. These include permeable reactive barriers (PRBs), bioremediation approaches, membrane filtration systems, and data-driven policy tools that address both immediate risks and long-term planning objectives. The framework explicitly incorporates enabling factors—such as multisectoral collaboration, funding availability, and incorporation of local knowledge—which support the implementation and sustainability of monitoring efforts. At the same time, it accounts for key barriers, including fragmented governance, infrastructure limitations, and data scarcity, which often constrain system performance and scalability in urban areas of the Philippines. Overall, the framework adopts a systems approach by integrating technological, institutional, and environmental dimensions to enable adaptive, evidence-based groundwater management. Unlike conventional frameworks that emphasize only hydrogeological or engineering aspects, this model foregrounds context sensitivity, recognizing the role of governance, infrastructure readiness, and urban development dynamics in shaping practical outcomes. The inclusion of both enablers and barriers ensures the framework is grounded in real-world applicability and encourages the development of scalable, community-aligned solutions.

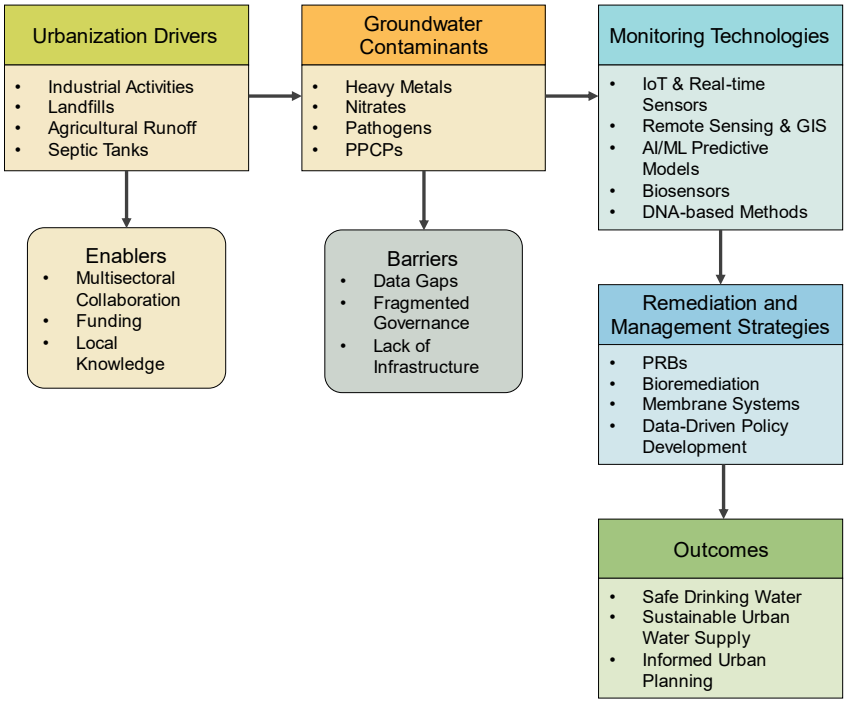


Figure 3. Conceptual framework for integrated urban groundwater monitoring and management in the Philippines, highlighting the interplay between urbanization pressures, contaminants, monitoring technologies, context-specific enablers and barriers, and remediation outcomes.

6.3. Determination of Research and Implementation Gaps

From the synthesis and conceptual model, several key research and implementation gaps have been identified. However, a fuller discussion is warranted here to elaborate on their implications and pathways forward.

First, the lack of AI-integrated real-time monitoring systems in Philippine settings undermines proactive contamination management. While AI models have demonstrated value in predicting water quality indices elsewhere, they are rarely applied alongside sensor networks in the country. This gap leads to delayed response times and missed opportunities for early intervention.

Second, biosensors and DNA-based techniques have great potential for detecting specific pollutants with high sensitivity, but they remain underutilized locally. Their limited deployment reflects both the novelty of the technologies and the lack of field validation in tropical, urbanized environments. This suggests a research opportunity to adapt and test these tools in representative Philippine conditions.

Third, the literature and existing projects disproportionately focus on legacy contaminants like heavy metals and nitrates. Emerging threats such as PPCPs and VOCs are severely under-researched despite increasing use of personal care products and industrial solvents. Without baseline data on these contaminants, regulatory frameworks cannot be designed effectively.

Fourth, a disconnection exists between monitoring outputs and their practical application in urban governance. Data collected by environmental agencies or researchers are seldom used to inform land-use planning or infrastructure development. As a result, new developments proceed in vulnerable aquifer zones without consideration for groundwater risks. Bridging this gap requires integrated planning tools, including zoning overlays informed by DRASTIC or machine-learning vulnerability models.

Fifth, institutional fragmentation limits the scale and effectiveness of groundwater protection. LGUs, water utilities, and national agencies often operate with different data protocols, budgets, and mandates. This results in duplicated efforts or inaction. A coordinated framework for data sharing and multisectoral governance is urgently needed.

These gaps are not merely technical—they reflect a systemic misalignment between environmental threats, technological potential, and governance readiness. Addressing them requires holistic strategies that cut across research, implementation, and policy reform.

6.4. Future Research and Policy Direction

Addressing the research gaps identified in the previous section demands an integrated set of strategies that combine technological innovation with governance reform. Moving forward, several priority areas can help transform groundwater monitoring and management practices in the Philippines.

One of the most pressing needs is the development of integrated platforms that combine IoT sensors with AI-based analytics. These systems can support real-time monitoring, predictive modeling, and early-warning mechanisms for contaminants such as nitrates, heavy metals, and pathogens. Pilot projects should be initiated in highly urbanized areas such as Metro Manila, Cebu, and Davao to demonstrate feasibility and guide future scaling.

Equally important is the field validation of emerging monitoring tools. Biosensors and DNA-based methods, while promising, must be tested under Philippine field

conditions to assess reliability, durability, and applicability. Research collaborations between academic institutions, government laboratories, and LGUs can facilitate these trials.

Expanding the scope of monitored contaminants is another critical direction. While heavy metals and nitrates are relatively well-documented, VOCs and PPCPs remain largely unexplored. These emerging pollutants require not only detection but also integration into regulatory frameworks and local urban planning decisions. National funding programs should support studies that establish baseline concentrations, evaluate potential health risks, and inform new regulatory thresholds. In addition, data from these studies should be channeled into real-time policy dashboards, zoning overlays, and infrastructure planning to ensure that scientific evidence informs actual urban management decisions.

Furthermore, the integration of groundwater monitoring into land-use planning and zoning is essential for long-term protection. Environmental impact assessments (EIAs) should mandate the use of aquifer vulnerability maps and risk overlays informed by geospatial and machine-learning models. This will help prevent high-risk developments in sensitive recharge areas.

Institutional reforms are also necessary to ensure sustained and coordinated action. A centralized groundwater quality database should be developed to allow for interagency data sharing and harmonized monitoring standards. Agreements among the Department of Environment and Natural Resources (DENR), LGUs, academic institutions, and water utilities can establish shared responsibilities and protocols to support integrated monitoring and rapid decision-making.

Lastly, community-based monitoring can offer a bottom-up complement to formal programs. Providing low-cost sensors and mobile applications to local communities can democratize data collection, enhance transparency, and foster public engagement. These initiatives also present an opportunity to foster citizen science networks, where community-generated data can be validated and integrated into municipal-level decision-making platforms, closing the feedback loop between monitoring and governance.

These directions are summarized in Table 8, which links each identified gap with its broader implications and proposed solutions:

Table 8. Summary of research and implementation gaps.

Identified Gap	Why It Matters	Consequence	Suggested Direction
No AI-integrated real-time monitoring	Limits early detection of pollution	Delayed interventions	Develop unified AI-IoT sensor platforms
Lack of local biosensor validation	Tools may not work in tropical urban contexts	Reduced trust, wasted funds	Field validation in PH cities
Understudied emerging contaminants (PPCPs, VOCs)	Cannot regulate what is not understood	Public health risk	Fund studies on PPCPs in Metro Manila/Cebu
Monitoring not linked to zoning or permits	Developments proceed in vulnerable areas	Long-term groundwater degradation	Integrate DRASTIC/ML maps in urban planning
Fragmented institutional coordination	Redundancy or inaction in groundwater protection	Governance inefficiency	Create multisector data-sharing and response hub

These strategies collectively aim to strengthen the Philippines' ability to safeguard its urban groundwater resources through localized, technology-driven, and participatory approaches.

7. Conclusions

The growing complexity of groundwater contamination in heavily urbanized areas, particularly in the Philippines, underscores the urgent need for integrated monitoring and

remediation strategies. This review reveals that while numerous technological solutions exist globally—ranging from IoT-based systems and biosensors to AI-driven models—their practical implementation in the local context remains limited. Effective groundwater protection demands not only the adoption of advanced tools but also improved institutional coordination, site-specific validation, and the seamless integration of scientific data into planning and policy-making processes.

Technological innovation has expanded the arsenal of tools available for groundwater monitoring. IoT-based sensors, remote sensing techniques, machine-learning models, biosensors, and DNA-based methods significantly enhance the speed, accuracy, and predictive capacity of monitoring. However, their use in the Philippines is still limited due to high implementation costs, insufficient technical capacity, and fragmented institutional uptake. At the same time, urban groundwater systems are threatened by a diverse and evolving mix of contaminants—ranging from heavy metals and nitrates to pathogens, hydrocarbons, VOCs, and pharmaceuticals and personal care products (PPCPs)—each requiring tailored detection and remediation strategies. While detection methods are generally well-established, remediation is hindered by the lack of integrated frameworks and inconsistent deployment of technologies.

Moreover, the disconnect between data and policy continues to pose a significant barrier. Although tools such as GIS, DRASTIC vulnerability mapping, and AI-enabled forecasting are available, they are rarely utilized to inform zoning regulations, infrastructure design, or emergency response mechanisms. Bridging this divide will require robust policy mandates that embed environmental monitoring outputs into land-use planning and regulatory processes. Critical regulatory and monitoring gaps—such as the absence of baseline data on emerging contaminants, overlapping institutional responsibilities, and weak data integration into governance—further limit the responsiveness and effectiveness of local government units and regulatory bodies.

To address these challenges, future research should prioritize the local validation of technologies, integrated monitoring approaches for multiple contaminants, and the development of governance models that support coordinated and inclusive decision-making. Pilot studies must focus on adapting technologies to Philippine urban conditions, evaluating contaminant interactions in mixed-pollution scenarios, and promoting participatory governance frameworks. These should include community-based monitoring initiatives, inter-agency collaboration, and the integration of AI and IoT systems to enable real-time decision support in groundwater management.

Author Contributions: Conceptualization, K.P.V.R. and C.E.F.M.; methodology, K.P.V.R.; formal analysis, K.P.V.R.; investigation, C.E.F.M.; resources, K.P.V.R. and C.E.F.M.; data curation, K.P.V.R.; writing—original draft preparation, K.P.V.R.; writing—review and editing, C.E.F.M.; visualization, K.P.V.R. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Conflicts of Interest: The authors declare no conflicts of interest.

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